Customer Shopping Behavior Analysis

A data-driven project analyzing 3,900 transactions to uncover spending patterns, customer segments, product preferences, and subscription behavior.

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Project Overview

What We Set Out to Do

Analyze transactional data from **3,900 purchases** across product categories to guide strategic business decisions.

Spending Patterns

Customer Segments

Product Preferences

Subscription Behavior

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Dataset Summary

3,900

Rows

Total purchase records

18

Columns

Features per record

37

Missing Values

In Review Rating column

Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Behavior

Discounts, Promo Codes, Frequency, Ratings, Shipping

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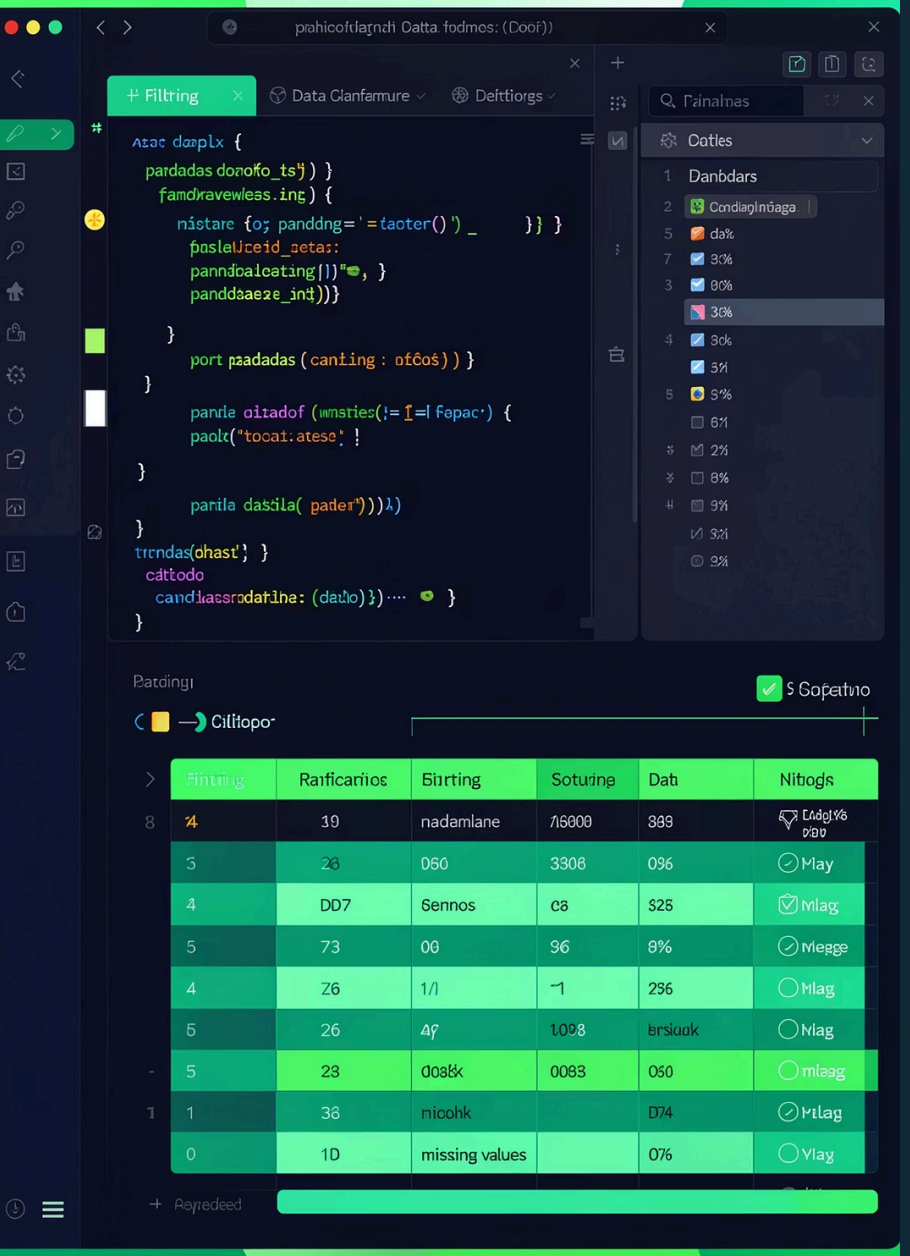

Exploratory Data Analysis in Python

Data was loaded with **pandas**, explored with df.info() and .describe(), and cleaned for analysis.

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Data Preparation & Feature Engineering

Load & Explore

Import data and inspect distributions

Clean & Impute

Handle missing values; median by category

Engineer Features

Create age_group and purchase_frequency_days

Export to PostgreSQL

Save cleaned table to production DB

Missing Review Ratings were imputed using **median per product category**. New columns included **age_group** and **purchase_frequency_days**. Redundant `promo_code_used` column was dropped after consistency checks.

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SQL Analysis: Revenue & Customer Insights

Revenue by Gender

Compared total revenue from male vs. female customers.

High-Spending Discount Users

Identified customers using discounts yet spending above average.

Revenue by Age Group

Calculated total revenue contribution of each age group.

Subscribers vs. Non-Subscribers

Compared average spend and total revenue by subscription status.

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SQL Analysis: Products & Shipping

Top 5 Products by Rating

Found products with the highest average review ratings.

Top 3 Products per Category

Listed the most purchased products within each category.

Discount-Dependent Products

Identified 5 products with the highest % of discounted purchases.

Shipping Type Comparison

Compared average purchase amounts: Standard vs. Express shipping.

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Customer Segmentation

Segmentation Logic

Customers were classified using **purchase history** in PostgreSQL:

- **New** – First-time buyers
- **Returning** – Moderate purchase history
- **Loyal** – 5+ purchases; significantly more likely to subscribe

ⓘ Repeat buyers with 5+ purchases showed a strong correlation with active subscriptions.

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Power BI Dashboard

An interactive dashboard was built in **Power BI** to present all key insights visually — enabling stakeholders to explore spending patterns, segment performance, and product trends at a glance.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to drive subscriber growth.

Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

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